$\therefore$ The parallel and perpendicular lines are respectively

$$\begin{aligned} 3x + 4y &= 11 \\ 4x - 3y &= -2 \end{aligned}$$

Example 6.23 Find the distance

(i) between two points (5, 4) and (2, 0)
(ii) from a point (1, 2) to the line $5x + 12y - 3 = 0$
(iii) between two parallel lines $3x + 4y = 12$ and $6x + 8y + 1 = 0$.

Solution:

(i) Distance between two points $(x_1 y_1) = (5, 4)$ and $(x_2, y_2) = (2, 0)$ is

$$\begin{aligned} D &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \\ &= \sqrt{3^2 + 4^2} \\ D &= 5 \end{aligned}$$

(ii) Distance between the point (x_1, y_1) and the line $ax + by + c = 0$ is

$$D = \left| \frac{ax_1 + by_1 + c}{\sqrt{a^2 + b^2}} \right|$$

$\therefore$ The distance between the point (1, 2) and the line $5x + 12y - 3 = 0$ is

$$\begin{aligned} D &= \left| \frac{5(1) + 12(2) - 3}{\sqrt{5^2 + 12^2}} \right| \\ D &= 2 \end{aligned}$$

(iii) Distance between two parallel lines $a_1x + b_1y + c_1 = 0$ and $a_1x + b_1y + c_2 = 0$ is

$$D = \frac{|c_1 - c_2|}{\sqrt{a_1^2 + b_1^2}}$$

Given lines can be written as $3x + 4y - 12 = 0$ and $3x + 4y + \frac{1}{2} = 0$
Here $a_1 = 3, b_1 = 4, c_1 = -12, c_2 = \frac{1}{2}$

$$D = \frac{|c_1 - c_2|}{\sqrt{a_2^1 + b_2^1}} = \left| \frac{-12 - \frac{1}{2}}{\sqrt{3^2 + 4^2}} \right| = \frac{25}{2 \times 5} = 2.5 \text{ units.}$$

Example 6.24 Find the nearest point on the line $2x + y = 5$ from the origin.

Solution:

The required point is the foot of the perpendicular from the origin on the line $2x + y = 5$.
The line perpendicular to the given line, through the origin is $x - 2y = 0$.
Solving the equations $2x + y = 5$ and $x - 2y = 0$, we get $x = 2, y = 1$
Hence the nearest point on the line from the origin is (2, 1).